

Zero-Density Concentration for Dirichlet Polynomials

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Abstract

We prove that broad families of finite Dirichlet sums exhibit deterministic concentration of normalized vertical zero density on a single line. The main result gives a general criterion under which the normalized Jessen potentials converge locally uniformly to a piecewise-linear convex function with a single corner. The associated normalized measures describing the distribution of real parts of zeros in vertical mean density then converge weakly to a Dirac mass. Thus, for each fixed truncation, zeros may occupy a nontrivial range of real parts, while asymptotically their normalized vertical density concentrates on one line.

The proof combines the finite Bohr lift with a translation-uniform anti-concentration estimate for isolated prime coordinates. We apply the criterion to partial sums of the Riemann zeta function, fixed Dirichlet L -functions, and primitive holomorphic Hecke eigenforms of fixed level, trivial Dirichlet character, and without complex multiplication. The concentration line is $\operatorname{Re} s = 1/2$ in the normalized setting and becomes $\operatorname{Re} s = k/2$ for the classical Fourier coefficients of a form of weight k .

Keywords. Dirichlet polynomials; zeros of partial L -functions; zero-density concentration; Jessen function; almost periodic functions; Bohr lift; Mahler measure; anti-concentration; Steinhaus sums; Sato–Tate; Rankin–Selberg.

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1 Introduction

Let

$$P_N(s) = \sum_{n \leq N} a_n n^{-s}, \quad s = \sigma + it, \quad (1)$$

be a finite Dirichlet sum with $a_1 \neq 0$. For each fixed N , P_N is an exponential polynomial in the vertical variable. Its zeros may occupy a wide range of real parts, and a substantial literature has developed around their location, existence, density, and zero-free regions.

For the special case

$$\zeta_N(s) = \sum_{n \leq N} n^{-s}, \quad (2)$$

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Montgomery studied the extreme right-hand zeros [17]; Borwein, Fee, Ferguson and van der Waall gave extensive numerical and structural information [4]; Gonek and Ledoan obtained zero-counting results and information on the distribution of real parts in finite-height rectangles [9]; and Mora proved that, for every sufficiently large fixed N , the set of real parts of the zeros is dense throughout the corresponding critical strip [18]. These results show in particular that the support of the zero set of a fixed truncation can be much wider than any single distinguished vertical line.

The same circle of questions has been pursued for broader arithmetic Dirichlet polynomials. Ledoan, Roy and Zaharescu studied zero-free regions and zero counts for partial sums of Dedekind zeta functions of cyclotomic fields [15]. Roy and Vatwani developed zero-free regions and zero-density estimates for partial sums attached to multiplicative functions [21], and later proved sharp existence results for zeros of the corresponding Dirichlet polynomials [22]. In a complementary direction, the author used Bohr equivalence to prove density of attainable real parts for partial sums of the Dirichlet lambda, beta and eta functions [7]. Very recently, Kerr, Klurman and Thorner [14] proved that unusually large partial sums of Dirichlet coefficients can force low-lying zeros of the underlying infinite L -function away from the critical line. Their zeros are zeros of $L(s, \pi)$ itself, whereas the present paper concerns zeros of the finite Dirichlet truncations.

There is also related work for modular forms with a different approximating object. Li, Roy and Zaharescu considered symmetrized approximations to Hecke L -functions obtained by adjoining to a truncated Dirichlet series the dual term supplied by the functional equation [16]. Their approximants are structurally different from the raw truncations considered here, and their asymptotic regime couples the truncation parameter with the height.

The question addressed in this paper is instead the following: *where does the normalized vertical density of zeros concentrate as the truncation length tends to infinity?* The distinction between support and mass is essential. A family can have zeros with real parts dense throughout a large strip while an asymptotically negligible proportion of its vertical zero density lies away from one distinguished line.

For fixed N , the relevant convex potential is the Jessen function

$$J_{P_N}(\sigma) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \log |P_N(\sigma + it)| dt. \quad (3)$$

The existence of this mean follows from Jessen and Tornehave [12, Theorem 5], while its convexity follows from [12, Theorem 7]. Their Jensen formula for ordinary Dirichlet series [12, Theorem 31] identifies $\frac{1}{2\pi} J_{P_N}''$, in the sense of distributions, with the vertical zero-frequency measure. Recent work of Andersson [1] develops related mean-counting formulae for vertical limits of infinite Dirichlet series. Our setting is finite-dimensional and introduces the asymptotic parameter $N \rightarrow \infty$.

The novelty here is a limit theorem for the whole Jessen potential. Under an explicit isolated-prime hypothesis, the normalized potentials converge locally uniformly to a piecewise-linear function with one corner; the corresponding normalized zero-density measures converge to a single Dirac mass. For zeta and fixed Dirichlet L -functions the limiting line is $\operatorname{Re} s = 1/2$. The same line occurs for normalized holomorphic Hecke eigenvalues, while the classical Fourier-coefficient normalization moves it to $\operatorname{Re} s = k/2$.

The principal analytic input is a translation-uniform logarithmic anti-concentration estimate for sums of independent Steinhaus variables with comparable weights. More precisely, if $Z_1, \dots, Z_m$ are independent Steinhaus variables and $b_1, \dots, b_m > 0$ satisfy $\max_j b_j / \min_j b_j \leq K$, then, uni-

formly in $a \in \mathbb{C}$,

$$\mathbb{E} \left(\log \left| a + \sum_{j=1}^m b_j Z_j \right| \right) \geq \frac{1}{2} \log \sum_{j=1}^m b_j^2 - C_K.$$

The uniformity in the translation is essential: after conditioning on the remaining coordinates of the finite Bohr torus, the isolated prime-coordinate block is translated by a complex value depending on those conditioned coordinates.

All density statements below use the Jessen mean-motion order of limits: for each fixed N the height T tends to infinity first, and only afterwards $N \rightarrow \infty$. No coupled (N, T) limit is asserted. This is different from the finite-height regime studied, for example, by Gonek and Ledoan [9].

The paper is organized as follows. Section 2 states the abstract zero-density concentration criterion and its three arithmetic applications. Sections 3–5 develop the Bohr–Jessen framework, prove the translation-uniform anti-concentration estimate, and establish the abstract criterion. Sections 6–8 verify its hypotheses for zeta partial sums, fixed Dirichlet L -functions, and holomorphic Hecke eigenforms, respectively.

2 Main results

Let

$$M_N := \max\{n \leq N : a_n \neq 0\}. \quad (4)$$

When $a_1 \neq 0$, Proposition 3.7 below shows that J''_{P_N} has total mass $\log M_N$. Recall also that a finite convex function on an open interval has monotone one-sided derivatives [20, Theorem 24.1]; consequently its distributional second derivative is a positive Radon measure, namely the Lebesgue–Stieltjes measure associated with its increasing derivative.

Definition 2.1 (Zero-density concentration). Assume $M_N \rightarrow \infty$. We say that the family $\{P_N\}$ exhibits *zero-density concentration* on the line $\operatorname{Re} s = \alpha$ if

$$\nu_N := \frac{1}{\log M_N} J''_{P_N} \xrightarrow{w} \delta_\alpha \quad (5)$$

weakly as probability measures on $\mathbb{R}$.

This definition has a direct zero-counting interpretation. By Jessen and Tornehave [12, Theorem 31], for every $a < b$,

$$\frac{J''_{P_N}((a, b))}{2\pi} = \lim_{T \rightarrow \infty} \frac{1}{2T} \#\{\rho = \beta + i\gamma : P_N(\rho) = 0, a < \beta < b, |\gamma| < T\}, \quad (6)$$

where zeros are counted with multiplicity. Thus (5) means that asymptotically all normalized vertical zero density lies in every fixed neighborhood of the line $\operatorname{Re} s = \alpha$.

For P_N as in (1), normalize $a_1 = 1$ and set

$$E_N(\sigma) := \sum_{n \leq N} |a_n|^2 n^{-2\sigma}. \quad (7)$$

Theorem 2.2 (Isolated-prime zero-density criterion). *Let $P_N(s) = \sum_{n \leq N} a_n n^{-s}$ with $a_1 = 1$, and let M_N be given by (4). Suppose there exist $\alpha \in \mathbb{R}$ and subsets*

$$\mathcal{Q}_N \subset \{p \text{ prime} : N/2 < p \leq N\}$$

with $\#\mathcal{Q}_N \rightarrow \infty$ such that:

(H1) for every $\sigma \in \mathbb{R}$,

$$\frac{1}{2 \log N} \log E_N(\sigma) \longrightarrow (\alpha - \sigma)_+; \quad (8)$$

(H2) for every compact interval $I \subset (-\infty, \alpha)$ there is $K_I \geq 1$ such that, for all sufficiently large N , all $\sigma \in I$, and all $p, q \in \mathcal{Q}_N$,

$$K_I^{-1} \leq \frac{|a_p| p^{-\sigma}}{|a_q| q^{-\sigma}} \leq K_I, \quad (9)$$

and, with

$$B_N(\sigma) := \sum_{p \in \mathcal{Q}_N} |a_p|^2 p^{-2\sigma}, \quad (10)$$

one has

$$\frac{1}{2 \log N} \log B_N(\sigma) \longrightarrow \alpha - \sigma \quad (11)$$

locally uniformly on $(-\infty, \alpha)$.

Then

$$\frac{J_{P_N}(\sigma)}{\log N} \longrightarrow (\alpha - \sigma)_+ \quad (12)$$

locally uniformly on $\mathbb{R}$, and

$$\frac{1}{\log M_N} J_{P_N}'' \xrightarrow{w} \delta_\alpha. \quad (13)$$

In particular, the normalized vertical zero density concentrates on the line $\operatorname{Re} s = \alpha$.

Remark 2.3. The criterion is not intrinsically restricted to the arithmetic families considered below. Its application to further families reduces to two arithmetic inputs: the quadratic-energy asymptotic in hypothesis (H1), and a sufficiently large family of isolated primes on which the normalized prime coefficients are bounded above and away from zero, as required in hypothesis (H2). A suitable Sato–Tate-type equidistribution theorem provides a natural mechanism for the latter condition.

Theorem 2.4 (Riemann zeta partial sums). *For*

$$\zeta_N(s) = \sum_{n \leq N} n^{-s},$$

let J_N denote its Jessen function. Then

$$\frac{J_N(\sigma)}{\log N} \longrightarrow \left(\frac{1}{2} - \sigma \right)_+ \quad (14)$$

locally uniformly on $\mathbb{R}$, and

$$\frac{1}{\log N} J_N'' \xrightarrow{w} \delta_{1/2}. \quad (15)$$

Theorem 2.5 (Dirichlet L -functions). *Let χ be a fixed Dirichlet character modulo q , and put*

$$L_N(s, \chi) = \sum_{n \leq N} \chi(n) n^{-s}.$$

Let $J_{N,\chi}$ be its Jessen function and

$$M_{N,\chi} = \max\{n \leq N : (n, q) = 1\}.$$

Then

$$\frac{J_{N,\chi}(\sigma)}{\log N} \longrightarrow \left(\frac{1}{2} - \sigma\right)_+ \quad (16)$$

locally uniformly on $\mathbb{R}$, and

$$\frac{1}{\log M_{N,\chi}} J''_{N,\chi} \xrightarrow{w} \delta_{1/2}. \quad (17)$$

For the modular application, let f be a fixed normalized primitive holomorphic Hecke eigenform of even weight $k \geq 2$, fixed level Q , and trivial Dirichlet character; see, for example, Iwaniec [11, Sections 6.6–6.8] for the theory and normalization of newforms. Thus f has a Fourier expansion at the cusp infinity of the form

$$f(z) = \sum_{n \geq 1} a_f(n) e^{2\pi i n z}, \quad a_f(1) = 1, \quad (18)$$

where $a_f(n)$ denotes the n th Fourier coefficient of f .

We further assume that f has no complex multiplication. This is the hypothesis under which the Sato–Tate equidistribution theorem used below applies; see Barnet-Lamb, Gee and Geraghty [2, Corollary 7.1.7].

Define the normalized Hecke eigenvalues by

$$\lambda_f(n) := a_f(n) n^{-(k-1)/2}. \quad (19)$$

With this normalization, the associated L -function is centered at $\operatorname{Re} s = 1/2$.

Theorem 2.6 (Fixed-level holomorphic Hecke eigenforms). *With f as above, define*

$$L_N(s, f) = \sum_{n \leq N} \lambda_f(n) n^{-s}. \quad (20)$$

Then the Jessen functions $J_{N,f}$ satisfy

$$\frac{J_{N,f}(\sigma)}{\log N} \longrightarrow \left(\frac{1}{2} - \sigma\right)_+ \quad (21)$$

locally uniformly on $\mathbb{R}$, and

$$\frac{1}{\log M_{N,f}} J''_{N,f} \xrightarrow{w} \delta_{1/2}, \quad (22)$$

where $M_{N,f} := \max\{n \leq N : \lambda_f(n) \neq 0\}$.

Equivalently, for the classical Fourier-coefficient truncations

$$\mathcal{L}_N(s, f) = \sum_{n \leq N} a_f(n) n^{-s}, \quad (23)$$

zero-density concentration occurs on the line

$$\operatorname{Re} s = \frac{k}{2}. \quad (24)$$

3 Bohr–Jessen framework and zero-density measures

3.1 Finite Bohr lifts

Fix $N \geq 2$, let $p_1 < \dots < p_d \leq N$ be the primes up to N , $d = \pi(N)$ and write

$$n = \prod_{j=1}^d p_j^{v_{p_j}(n)}, \quad \mathbf{v}(n) := (v_{p_1}(n), \dots, v_{p_d}(n)) \in \mathbb{N}_0^d. \quad (25)$$

For $z = (z_1, \dots, z_d) \in \mathbb{T}^d$, define the Bohr lift

$$F_{P_N}(\sigma, z) := \sum_{n \leq N} a_n n^{-\sigma} z^{\mathbf{v}(n)}, \quad z^{\mathbf{m}} := z_1^{m_1} \dots z_d^{m_d}. \quad (26)$$

Then

$$F_{P_N}(\sigma, (p_1^{-it}, \dots, p_d^{-it})) = P_N(\sigma + it). \quad (27)$$

The elementary arithmetic fact behind the full-torus model is worth isolating.

Proposition 3.1 (Prime-frequency injectivity). *Let*

$$\omega_N = (\log p_1, \dots, \log p_d).$$

Then the resonance lattice

$$\Lambda_N := \{\mathbf{k} \in \mathbb{Z}^d : \langle \mathbf{k}, \omega_N \rangle = 0\}$$

is trivial. Consequently the Kronecker orbit

$$t \mapsto (p_1^{-it}, \dots, p_d^{-it})$$

is dense in $\mathbb{T}^d$, and the map $n \mapsto \mathbf{v}(n)$ is injective.

Proof. If $\mathbf{k} = (k_1, \dots, k_d) \in \Lambda_N$, then

$$0 = \sum_{j=1}^d k_j \log p_j = \log \left(\prod_{j=1}^d p_j^{k_j} \right).$$

Hence $\prod_j p_j^{k_j} = 1$. Unique factorization in $\mathbb{Z}$ forces $k_j = 0$ for every j , so $\Lambda_N = \{0\}$. The continuous Kronecker–Weyl theorem then implies that the orbit is uniformly distributed, and in particular dense, in the full torus; see Haviland and Wintner [10]. Jessen and Tornehave use this finite-integral-base torus representation in their spatial-extension construction [12, §116]. Finally, $\mathbf{v}(m) = \mathbf{v}(n)$ implies $m = n$ by unique factorization. $\square$

3.2 The exact Bohr–Jessen identity

At a zero of $F_{P_N}(\sigma, \cdot)$ one has $\log |F_{P_N}| = -\infty$. Thus the logarithm is neither continuous nor bounded on the torus, so the continuous Kronecker–Weyl theorem invoked above cannot be applied to it directly. We first record the required logarithmic integrability.

Lemma 3.2. *If Q is a nonzero complex polynomial in d variables, then*

$$\log |Q| \in L^1(\mathbb{T}^d).$$

Proof. This is the standard finiteness statement underlying logarithmic Mahler measure; compare Boyd [3, Section 1]. We include the argument because the integrability near the zero set will be used below. Multiplication by a monomial does not change the modulus on $\mathbb{T}^d$, so it is enough to consider an ordinary polynomial.

For $d = 1$, write

$$Q(z) = c \prod_{j=1}^r (z - \alpha_j).$$

Since Q is continuous on $\mathbb{T}$, the positive part $\log^+ |Q|$ is bounded. It remains to check each factor in the negative part.

If $|\alpha| \neq 1$, then

$$\inf_{\theta \in [0, 2\pi]} |e^{i\theta} - \alpha| > 0,$$

so $\log |e^{i\theta} - \alpha|$ is bounded.

If $|\alpha| = 1$, write $\alpha = e^{i\theta_0}$. Then

$$|e^{i\theta} - e^{i\theta_0}| = 2 \left| \sin \frac{\theta - \theta_0}{2} \right|. \quad (28)$$

For $|\theta - \theta_0| \leq 1$, the elementary bounds $c_1|u| \leq |\sin u| \leq |u|$ give

$$c_2|\theta - \theta_0| \leq |e^{i\theta} - e^{i\theta_0}| \leq |\theta - \theta_0|.$$

Consequently the only possible singularity is logarithmic, and

$$\int_0^\delta |\log u| du = \delta(1 - \log \delta) < \infty \quad (0 < \delta < 1). \quad (29)$$

Hence $\log |Q| \in L^1(\mathbb{T})$.

Assume now that the assertion holds in $d - 1$ variables. Regard Q as a polynomial in the last variable z_d , with coefficients depending polynomially on the remaining variables $z' = (z_1, \dots, z_{d-1})$:

$$Q(z', z_d) = \sum_{j=0}^m Q_j(z') z_d^j, \quad z' \in \mathbb{T}^{d-1},$$

where $m = \deg_{z_d} Q$, each $Q_j \in \mathbb{C}[z_1, \dots, z_{d-1}]$, and $Q_m \not\equiv 0$. By the induction hypothesis, $\log |Q_m| \in L^1(\mathbb{T}^{d-1})$; in particular the zero set of Q_m has Haar measure zero. For every z' outside that null set, factor

$$Q(z', z_d) = Q_m(z') \prod_{j=1}^m (z_d - \alpha_j(z')).$$

Jensen's one-variable formula, in the form recorded for example in Conway [5, Chapter XI, Section 1], gives

$$\begin{aligned} L(z') &:= \int_{\mathbb{T}} \log |Q(z', z_d)| dm(z_d) \\ &= \log |Q_m(z')| + \sum_{j=1}^m \log^+ |\alpha_j(z')| \geq \log |Q_m(z')|. \end{aligned} \quad (30)$$

On the other hand, if $M := \max_{\mathbb{T}^d} |Q|$, then

$$P(z') := \int_{\mathbb{T}} \log^+ |Q(z', z_d)| \, dm(z_d) \leq \log^+ M.$$

Since $L = P - N$, where

$$N(z') := \int_{\mathbb{T}} \log^- |Q(z', z_d)| \, dm(z_d),$$

we obtain from (30)

$$0 \leq N(z') = P(z') - L(z') \leq \log^+ M + \log^- |Q_m(z')|.$$

The right-hand side is integrable on $\mathbb{T}^{d-1}$ by the induction hypothesis. Since $\log^+ |Q|$ is nonnegative, Tonelli's theorem gives

$$\begin{aligned} \int_{\mathbb{T}^d} \log^+ |Q(z)| \, dm_d(z) &= \int_{\mathbb{T}^{d-1}} \left(\int_{\mathbb{T}} \log^+ |Q(z', z_d)| \, dm(z_d) \right) dm_{d-1}(z') \\ &= \int_{\mathbb{T}^{d-1}} P(z') \, dm_{d-1}(z') \\ &\leq \log^+ M < \infty. \end{aligned}$$

Similarly, Tonelli's theorem applied to $\log^- |Q|$ yields

$$\begin{aligned} \int_{\mathbb{T}^d} \log^- |Q(z)| \, dm_d(z) &= \int_{\mathbb{T}^{d-1}} N(z') \, dm_{d-1}(z') \\ &\leq \log^+ M + \int_{\mathbb{T}^{d-1}} \log^- |Q_m(z')| \, dm_{d-1}(z') \\ &< \infty, \end{aligned}$$

where the last inequality follows from the induction hypothesis. Therefore

$$\begin{aligned} \int_{\mathbb{T}^d} |\log |Q(z)|| \, dm_d(z) &= \int_{\mathbb{T}^d} \log^+ |Q(z)| \, dm_d(z) + \int_{\mathbb{T}^d} \log^- |Q(z)| \, dm_d(z) \\ &< \infty. \end{aligned}$$

This completes the induction. □

Proposition 3.3 (Bohr–Jessen identity). *For every fixed N and every $\sigma \in \mathbb{R}$, the limit*

$$J_{P_N}(\sigma) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \log |P_N(\sigma + it)| \, dt \quad (31)$$

exists and

$$J_{P_N}(\sigma) = \int_{\mathbb{T}^d} \log |F_{P_N}(\sigma, z)| \, dm_d(z), \quad (32)$$

where m_d is normalized Haar measure on $\mathbb{T}^d$. Moreover J_{P_N} is finite and convex on $\mathbb{R}$.

Proof. The frequencies $-\log n$ are integral combinations of the finite basis $-\log p_1, \dots, -\log p_d$. Jessen and Tornehave [12, Theorem 5] show that, for a nonzero analytic almost periodic function, the logarithmic mean exists and is obtained by first truncating the logarithm from below and then letting the truncation level tend to zero.

Fix $\varepsilon > 0$ and set

$$g_{\sigma,\varepsilon}(z) := \log \max\{|F_{P_N}(\sigma, z)|, \varepsilon\}, \quad z \in \mathbb{T}^d.$$

This is continuous on $\mathbb{T}^d$. Define

$$\Phi_N : \mathbb{R} \longrightarrow \mathbb{T}^d, \quad \Phi_N(t) := (p_1^{-it}, \dots, p_d^{-it}).$$

By (27),

$$g_{\sigma,\varepsilon}(\Phi_N(t)) = \log \max\{|P_N(\sigma + it)|, \varepsilon\}.$$

Proposition 3.1 and the Kronecker–Weyl theorem therefore give

$$\begin{aligned} & \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \log \max\{|P_N(\sigma + it)|, \varepsilon\} dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T g_{\sigma,\varepsilon}(\Phi_N(t)) dt \\ &= \int_{\mathbb{T}^d} g_{\sigma,\varepsilon}(z) dm_d(z) \\ &= \int_{\mathbb{T}^d} \log \max\{|F_{P_N}(\sigma, z)|, \varepsilon\} dm_d(z). \end{aligned} \tag{33}$$

This is the finite-integral-base spatial extension used by Jessen and Tornehave [12, §116].

For $0 < \varepsilon \leq 1$,

$$|\log \max\{|F_{P_N}|, \varepsilon\}| \leq \log^+ \|F_{P_N}\|_\infty + \log^- |F_{P_N}|.$$

The right-hand side is integrable by Lemma 3.2. Hence dominated convergence gives

$$\lim_{\varepsilon \downarrow 0} \int_{\mathbb{T}^d} \log \max\{|F_{P_N}(\sigma, z)|, \varepsilon\} dm_d(z) = \int_{\mathbb{T}^d} \log |F_{P_N}(\sigma, z)| dm_d(z).$$

Combining this with (33) and [12, Theorem 5] proves (32). Finiteness follows from Lemma 3.2, while convexity follows from Jessen and Tornehave [12, Theorem 7]. $\square$

3.3 Jessen–Tornehave zero density

For a convex function J , write $J'(x-)$ and $J'(x+)$ for its left and right derivatives.

Proposition 3.4 (Vertical zero density). *Let $Z(P_N)$ denote the multiset of zeros $\rho = \beta + i\gamma$ of P_N . If $a < b$, then the vertical zero frequency in the strip $a < \operatorname{Re} s < b$ exists and*

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \#\{\rho \in Z(P_N) : a < \beta < b, |\gamma| < T\} = \frac{1}{2\pi} (J'_{P_N}(b-) - J'_{P_N}(a+)). \tag{34}$$

Equivalently,

$$\rho_{P_N} := \frac{1}{2\pi} J''_{P_N} \tag{35}$$

is the locally finite vertical zero-density measure on the real axis.

Proof. This is Jessen and Tornehave [12, Theorem 31], applied to the ordinary Dirichlet series P_N . Their Jensen function is

$$\varphi(\sigma) := \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \log |P_N(\sigma + it)| dt,$$

so in our notation $\varphi = J_{P_N}$. Theorem 31 gives, for every $a < b$,

$$H(a, b) = \frac{1}{2\pi}(\varphi'(b - 0) - \varphi'(a + 0)),$$

where $H(a, b)$ denotes the relative frequency of zeros in the strip $a < \operatorname{Re} s < b$. This is exactly (34). $\square$

Remark 3.5. For $f(s) = 1 + e^{-\lambda s}$,

$$J_f(\sigma) = \max\{0, -\lambda\sigma\}, \quad J_f'' = \lambda\delta_0.$$

The zeros are $s = (2m + 1)\pi i/\lambda$, whose vertical density in $|\operatorname{Im} s| < T$ is $\lambda/(2\pi)$. Thus (35) has the correct constant.

3.4 Elementary bounds and total mass

Proposition 3.6. *Assume $a_1 = 1$. Then for every $\sigma \in \mathbb{R}$,*

$$0 \leq J_{P_N}(\sigma) \leq \frac{1}{2} \log \sum_{n \leq N} |a_n|^2 n^{-2\sigma}. \quad (36)$$

Proof. For the upper bound, use Jensen's inequality for the concave function $x \mapsto \log x$: if μ is a probability measure and $X > 0$ is integrable, then

$$\int \log X \, d\mu \leq \log \int X \, d\mu.$$

Applying this to $X = |F_{P_N}(\sigma, z)|^2$ and normalized Haar measure gives

$$\begin{aligned} J_{P_N}(\sigma) &= \frac{1}{2} \int_{\mathbb{T}^d} \log |F_{P_N}(\sigma, z)|^2 \, dm_d(z) \\ &\leq \frac{1}{2} \log \int_{\mathbb{T}^d} |F_{P_N}(\sigma, z)|^2 \, dm_d(z). \end{aligned}$$

Distinct integers define distinct torus characters by Proposition 3.1. Their orthogonality therefore yields

$$\int_{\mathbb{T}^d} |F_{P_N}(\sigma, z)|^2 \, dm_d(z) = \sum_{n \leq N} |a_n|^2 n^{-2\sigma}.$$

For the lower bound, use Jensen's formula in one complex variable; see Conway [5, Chapter XI, Section 1]. If

$$Q(z) = c_0 + c_1 z + \cdots + c_m z^m = c_m \prod_{j=1}^m (z - \alpha_j),$$

then

$$\int_{\mathbb{T}} \log |Q(z)| \, dm(z) = \log |c_m| + \sum_{j=1}^m \log^+ |\alpha_j| \geq \log |c_0|,$$

where the last inequality is the equivalent form obtained from $|c_0| = |c_m| \prod_j |\alpha_j|$. Apply this successively to the variables $z_d, z_{d-1}, \dots, z_1$. Fubini is legitimate by Lemma 3.2. At each step the circle mean is bounded below by the logarithm of the modulus of the constant coefficient in

that variable, and after all variables have been integrated only the constant monomial remains. Hence

$$J_{P_N}(\sigma) = \int_{\mathbb{T}^d} \log |F_{P_N}(\sigma, z)| \, dm_d(z) \geq \log |a_1| = 0.$$

□

Proposition 3.7. *Assume $a_1 \neq 0$ and let M_N be defined by (4). Then*

$$J_{P_N}''(\mathbb{R}) = \log M_N, \quad \rho_{P_N}(\mathbb{R}) = \frac{\log M_N}{2\pi}. \quad (37)$$

Consequently $J_{P_N}'' / \log M_N$ is a probability measure whenever $M_N > 1$.

Proof. As $\sigma \rightarrow +\infty$,

$$F_{P_N}(\sigma, z) = a_1 + o(1)$$

uniformly on $\mathbb{T}^d$, and therefore

$$J_{P_N}(\sigma) = \log |a_1| + o(1).$$

As $\sigma \rightarrow -\infty$, the highest supported term dominates uniformly:

$$F_{P_N}(\sigma, z) = a_{M_N} M_N^{-\sigma} z^{\mathbf{v}(M_N)} (1 + o(1)),$$

whence

$$J_{P_N}(\sigma) = \log |a_{M_N}| - \sigma \log M_N + o(1).$$

Since J_{P_N} is convex, these asymptotics determine the limiting one-sided slopes. Indeed, the usual secant-slope inequalities for a convex function show that

$$J'_{P_N,+}(\sigma) \longrightarrow 0 \quad (\sigma \rightarrow +\infty),$$

and

$$J'_{P_N,+}(\sigma) \longrightarrow -\log M_N \quad (\sigma \rightarrow -\infty).$$

For a finite convex function, the one-sided derivatives are nondecreasing [20, Theorem 24.1]. Thus its distributional second derivative is the positive Lebesgue–Stieltjes measure associated with the increasing right derivative $J'_{P_N,+}$, characterized by

$$J_{P_N}''((a, b]) = J'_{P_N,+}(b) - J'_{P_N,+}(a);$$

see, for example, Folland [8, Section 1.5]. Letting $a \rightarrow -\infty$ and $b \rightarrow +\infty$ therefore gives

$$\begin{aligned} J_{P_N}''(\mathbb{R}) &= \lim_{b \rightarrow +\infty} J'_{P_N,+}(b) - \lim_{a \rightarrow -\infty} J'_{P_N,+}(a) \\ &= 0 - (-\log M_N) \\ &= \log M_N. \end{aligned}$$

This proves the first identity in (37). The second identity follows from Proposition 3.4:

$$\rho_{P_N}(\mathbb{R}) = \frac{1}{2\pi} J_{P_N}''(\mathbb{R}) = \frac{\log M_N}{2\pi}.$$

□

4 Translation-uniform anti-concentration

A *Steinhaus random variable* is, by definition, a random variable distributed according to normalized Haar measure on the unit circle $\mathbb{T}$; see, for example, Kahane [13, Chapter 1].

The Bessel function of the first kind of order zero is

$$J_0(u) := \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \left(\frac{u}{2}\right)^{2n}. \quad (38)$$

For real u it has the integral representation

$$J_0(u) = \frac{1}{2\pi} \int_0^{2\pi} e^{iu \cos \theta} d\theta, \quad (39)$$

and the classical large-argument expansion implies $J_0(u) = O(u^{-1/2})$ as $u \rightarrow +\infty$; see Watson [25, §§2.11, 2.2 and 7.21].

Lemma 4.1 (Translated logarithmic anti-concentration). *Fix $K \geq 1$. There exists $C_K < \infty$ such that the following holds for every $m \geq 5$. Let $Z_1, \dots, Z_m$ be independent Steinhaus variables and let $b_1, \dots, b_m > 0$ satisfy*

$$\frac{\max_j b_j}{\min_j b_j} \leq K. \quad (40)$$

Set

$$S^2 = \sum_{j=1}^m b_j^2. \quad (41)$$

Then, uniformly for every $a \in \mathbb{C}$,

$$\mathbb{E} \left(\log \left| a + \sum_{j=1}^m b_j Z_j \right| \right) \geq \log S - C_K. \quad (42)$$

Proof. Put

$$c_j = \frac{b_j}{S}, \quad X = \sum_{j=1}^m c_j Z_j.$$

Then $\sum_j c_j^2 = 1$. If $b_{\min} = \min_j b_j$ and $b_{\max} = \max_j b_j$, then

$$mb_{\min}^2 \leq S^2 \leq mb_{\max}^2,$$

and (40) gives

$$\frac{1}{K\sqrt{m}} \leq c_j \leq \frac{K}{\sqrt{m}} \quad (1 \leq j \leq m). \quad (43)$$

Identify $\mathbb{C}$ with $\mathbb{R}^2$ and let μ_X denote the probability law of X . We use the Fourier convention

$$\widehat{\mu}_X(\xi) := \int_{\mathbb{R}^2} e^{i\langle \xi, x \rangle} d\mu_X(x) = \mathbb{E} \left(e^{i\langle \xi, X \rangle} \right).$$

If $Z_j = e^{i\Theta_j}$ with Θ_j uniform on $[0, 2\pi)$, rotational invariance allows us to take the direction of ξ to be the first coordinate axis. Hence, by (39),

$$\begin{aligned}\mathbb{E}\left(e^{i\langle \xi, c_j Z_j \rangle}\right) &= \frac{1}{2\pi} \int_0^{2\pi} e^{ic_j |\xi| \cos \theta} d\theta \\ &= J_0(c_j |\xi|).\end{aligned}$$

Independence therefore gives

$$\widehat{\mu}_X(\xi) = \prod_{j=1}^m J_0(c_j |\xi|). \quad (44)$$

From the series (38),

$$J_0(u) = 1 - \frac{u^2}{4} + O(u^4) \quad (u \rightarrow 0).$$

Thus there exist absolute constants $u_0 \in (0, 1]$ and $c_0 > 0$ such that

$$|J_0(u)| \leq e^{-c_0 u^2} \quad (0 \leq u \leq u_0). \quad (45)$$

The large-argument expansion quoted above yields an absolute constant C_0 such that

$$|J_0(u)| \leq C_0 u^{-1/2} \quad (u \geq 1). \quad (46)$$

Moreover, (39) and the triangle inequality give

$$|J_0(u)| \leq \frac{1}{2\pi} \int_0^{2\pi} |e^{iu \cos \theta}| d\theta = 1.$$

If $u > 0$, equality would force $e^{iu \cos \theta}$ to have constant argument for almost every θ , which is impossible. Hence

$$|J_0(u)| < 1 \quad (u > 0). \quad (47)$$

Write $r = |\xi|$ and split the radial integral into three regions.

Region I: $0 \leq r \leq u_0 \sqrt{m}/K$. By (43), $c_j r \leq u_0$, so

$$|\widehat{\mu}_X(\xi)| \leq \exp\left(-c_0 r^2 \sum_{j=1}^m c_j^2\right) = e^{-c_0 r^2}. \quad (48)$$

Consequently

$$2\pi \int_0^{u_0 \sqrt{m}/K} r |\widehat{\mu}_X(r)| dr \leq 2\pi \int_0^\infty r e^{-c_0 r^2} dr = \frac{\pi}{c_0}. \quad (49)$$

Region II: $u_0 \sqrt{m}/K \leq r \leq RK \sqrt{m}$. Choose

$$R > \max\{1, u_0, C_0^2\}. \quad (50)$$

Then

$$\frac{u_0}{K^2} \leq c_j r \leq RK^2. \quad (51)$$

By continuity and (47),

$$\rho_K := \max_{u \in [u_0/K^2, RK^2]} |J_0(u)| < 1. \quad (52)$$

Therefore $|\widehat{\mu}_X(\xi)| \leq \rho_K^m$. Since the annulus has area at most $\pi R^2 K^2 m$,

$$\int_{u_0\sqrt{m}/K \leq |\xi| \leq RK\sqrt{m}} |\widehat{\mu}_X(\xi)| \, d\xi \leq \pi R^2 K^2 m \rho_K^m \leq C_K. \quad (53)$$

Region III: $r \geq RK\sqrt{m}$. Now $c_j r \geq R \geq 1$, so (46) and (43) yield

$$|\widehat{\mu}_X(\xi)| \leq \left(C_0 K^{1/2} m^{1/4} r^{-1/2} \right)^m. \quad (54)$$

For $m > 4$,

$$\begin{aligned} & \int_{RK\sqrt{m}}^{\infty} r \left(C_0 K^{1/2} m^{1/4} r^{-1/2} \right)^m \, dr \\ &= C_0^m K^{m/2} m^{m/4} \int_{RK\sqrt{m}}^{\infty} r^{1-m/2} \, dr \\ &= C_0^m K^{m/2} m^{m/4} \frac{(RK\sqrt{m})^{2-m/2}}{m/2 - 2} \\ &= \frac{K^2 m R^2}{m/2 - 2} \left(\frac{C_0^2}{R} \right)^{m/2}. \end{aligned} \quad (55)$$

For $m \geq 5$,

$$\frac{m}{m/2 - 2} = \frac{2m}{m - 4} \leq 10,$$

and $C_0^2/R < 1$. Thus the last expression is bounded uniformly in m , and multiplication by the polar factor 2π gives

$$\int_{|\xi| \geq RK\sqrt{m}} |\widehat{\mu}_X(\xi)| \, d\xi \leq C_K. \quad (56)$$

Combining the three regions,

$$\|\widehat{\mu}_X\|_{L^1(\mathbb{R}^2)} \leq C_K. \quad (57)$$

The Fourier inversion theorem for finite measures with integrable Fourier transform (see Rudin [23, Section 1.5.2]) now implies that μ_X is absolutely continuous with respect to planar Lebesgue measure. Its bounded continuous density is

$$f_X(x) := \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} e^{-i\langle \xi, x \rangle} \widehat{\mu}_X(\xi) \, d\xi, \quad (58)$$

so $d\mu_X(x) = f_X(x) \, dx$ and

$$\|f_X\|_{\infty} \leq (2\pi)^{-2} \|\widehat{\mu}_X\|_1 \leq C_K. \quad (59)$$

Consequently, uniformly in $w \in \mathbb{C}$ and $0 < r \leq 1$,

$$\begin{aligned} \mathbb{P}(|X + w| < r) &= \int_{\{x: |x+w| < r\}} f_X(x) \, dx \\ &\leq \pi r^2 \|f_X\|_{\infty} \leq C_K r^2. \end{aligned} \quad (60)$$

Set $Y := \log^- |X + w| \geq 0$. Pointwise,

$$Y = \int_0^{\infty} \mathbf{1}_{\{Y > t\}} \, dt.$$

Tonelli's theorem therefore gives the tail-integral formula

$$\begin{aligned}\mathbb{E}(Y) &= \int_0^\infty \mathbb{P}(Y > t) dt \\ &= \int_0^\infty \mathbb{P}(|X + w| < e^{-t}) dt \\ &\leq \int_0^\infty \min\{1, C_K e^{-2t}\} dt.\end{aligned}\tag{61}$$

Increasing C_K if necessary, we may assume $C_K \geq 1$. Put

$$t_K := \frac{1}{2} \log C_K.$$

Then $C_K e^{-2t} \geq 1$ exactly when $0 \leq t \leq t_K$, and therefore

$$\begin{aligned}\int_0^\infty \min\{1, C_K e^{-2t}\} dt &= \int_0^{t_K} 1 dt + C_K \int_{t_K}^\infty e^{-2t} dt \\ &= t_K + \frac{C_K}{2} e^{-2t_K} \\ &= \frac{1}{2} \log C_K + \frac{1}{2} =: C'_K.\end{aligned}\tag{62}$$

Hence

$$\mathbb{E}(Y) \leq C'_K.$$

Since $\log^+ |X + w| \geq 0$,

$$\mathbb{E}(\log |X + w|) \geq -C'_K.\tag{63}$$

Finally,

$$a + \sum_{j=1}^m b_j Z_j = S \left(\frac{a}{S} + X \right),$$

so (63) with $w = a/S$ proves (42). $\square$

Remark 4.2. The uniformity in the translation a is essential for the application below. After the other torus coordinates are fixed, the isolated prime-coordinate block is translated by a complex number depending on those fixed coordinates, and the estimate must remain uniform in that translation.

5 The abstract isolated-prime criterion

Let $P_N(s) = \sum_{n \leq N} a_n n^{-s}$ and let

$$\mathcal{Q}_N \subset \{p \text{ prime} : N/2 < p \leq N\}.$$

If $p \in \mathcal{Q}_N$ and $n \leq N$ is divisible by p , then $n = p$. Thus the Bohr coordinate z_p occurs only in the linear monomial $a_p p^{-\sigma} z_p$, and

$$F_{P_N}(\sigma, z) = A_N(\sigma, z') + \sum_{p \in \mathcal{Q}_N} a_p p^{-\sigma} z_p,\tag{64}$$

where A_N is independent of the coordinates indexed by $\mathcal{Q}_N$.

For a finite set $\mathcal{Q}$ of prime coordinates, write

$$dm_{\mathcal{Q}}(z) := \prod_{p \in \mathcal{Q}} dm(z_p),$$

where m is normalized Haar probability measure on $\mathbb{T}$. If $\mathcal{P}(N)$ denotes the set of all primes not exceeding N , set

$$\mathcal{R}_N := \mathcal{P}(N) \setminus \mathcal{Q}_N.$$

Then the Bohr torus decomposes as

$$\mathbb{T}^d = \mathbb{T}^{\mathcal{R}_N} \times \mathbb{T}^{\mathcal{Q}_N}, \quad dm_d = dm_{\mathcal{R}_N} dm_{\mathcal{Q}_N}.$$

Proposition 5.1 (Isolated-prime lower bound). *Let $I \subset \mathbb{R}$ be compact. Suppose that there exists $K_I \geq 1$ such that*

$$K_I^{-1} \leq \frac{|a_p|p^{-\sigma}}{|a_q|q^{-\sigma}} \leq K_I \quad (65)$$

for every $\sigma \in I$ and every $p, q \in \mathcal{Q}_N$, and suppose $\#\mathcal{Q}_N \geq 5$. Then

$$J_{P_N}(\sigma) \geq \frac{1}{2} \log \sum_{p \in \mathcal{Q}_N} |a_p|^2 p^{-2\sigma} - C_{K_I} \quad (\sigma \in I), \quad (66)$$

where C_{K_I} is the constant from Lemma 4.1.

Proof. Fix $\sigma \in I$ and fix all torus coordinates not indexed by $\mathcal{Q}_N$. Denote this collection by

$$z' = (z_p)_{p \in \mathcal{R}_N} \in \mathbb{T}^{\mathcal{R}_N}.$$

Then, by (64),

$$F_{P_N}(\sigma, z', z_{\mathcal{Q}_N}) = A_N(\sigma, z') + \sum_{p \in \mathcal{Q}_N} a_p p^{-\sigma} z_p,$$

where, for the fixed values of σ and z' , the quantity $A_N(\sigma, z') \in \mathbb{C}$ is independent of the coordinates z_p , $p \in \mathcal{Q}_N$.

Write

$$a_p = |a_p| e^{i\vartheta_p}, \quad p \in \mathcal{Q}_N.$$

Since normalized Haar measure on $\mathbb{T}$ is invariant under rotations, the changes of variables

$$z_p \mapsto e^{i\vartheta_p} z_p, \quad p \in \mathcal{Q}_N,$$

absorb the phases of the coefficients without changing the inner integral. Hence, for fixed σ and z' , Lemma 4.1 applies with the complex translation

$$a = A_N(\sigma, z')$$

and positive coefficients

$$b_p(\sigma) := |a_p| p^{-\sigma}, \quad p \in \mathcal{Q}_N.$$

By (65),

$$\frac{\max_{p \in \mathcal{Q}_N} b_p(\sigma)}{\min_{p \in \mathcal{Q}_N} b_p(\sigma)} \leq K_I.$$

Since $\#\mathcal{Q}_N \geq 5$, Lemma 4.1 therefore gives, uniformly in the translation $A_N(\sigma, z')$,

$$\begin{aligned} & \int_{\mathbb{T}^{\mathcal{Q}_N}} \log |F_{P_N}(\sigma, z', z_{\mathcal{Q}_N})| \, dm_{\mathcal{Q}_N}(z_{\mathcal{Q}_N}) \\ & \geq \frac{1}{2} \log \sum_{p \in \mathcal{Q}_N} |a_p|^2 p^{-2\sigma} - C_{K_I}. \end{aligned} \quad (67)$$

The right-hand side of (67) is independent of z' . Integrating over $\mathbb{T}^{\mathcal{R}_N}$ and using the product decomposition

$$dm_d = dm_{\mathcal{R}_N} dm_{\mathcal{Q}_N},$$

together with Proposition 3.3, we obtain

$$\begin{aligned} J_{P_N}(\sigma) &= \int_{\mathbb{T}^d} \log |F_{P_N}(\sigma, z)| \, dm_d(z) \\ &= \int_{\mathbb{T}^{\mathcal{R}_N}} \left[\int_{\mathbb{T}^{\mathcal{Q}_N}} \log |F_{P_N}(\sigma, z', z_{\mathcal{Q}_N})| \, dm_{\mathcal{Q}_N}(z_{\mathcal{Q}_N}) \right] dm_{\mathcal{R}_N}(z') \\ &\geq \int_{\mathbb{T}^{\mathcal{R}_N}} \left[\frac{1}{2} \log \sum_{p \in \mathcal{Q}_N} |a_p|^2 p^{-2\sigma} - C_{K_I} \right] dm_{\mathcal{R}_N}(z') \\ &= \frac{1}{2} \log \sum_{p \in \mathcal{Q}_N} |a_p|^2 p^{-2\sigma} - C_{K_I}, \end{aligned}$$

because $m_{\mathcal{R}_N}(\mathbb{T}^{\mathcal{R}_N}) = 1$. □

Proof of Theorem 2.2. We first record that hypothesis (H2) automatically determines the scale of M_N . For all sufficiently large N , the inequalities in (9) force $a_p \neq 0$ for every $p \in \mathcal{Q}_N$. Since $\#\mathcal{Q}_N \rightarrow \infty$, the set $\mathcal{Q}_N$ is nonempty for all sufficiently large N ; choosing any $p \in \mathcal{Q}_N$ gives

$$\frac{N}{2} < p \leq M_N \leq N.$$

Consequently

$$1 - \frac{\log 2}{\log N} < \frac{\log M_N}{\log N} \leq 1, \quad (68)$$

and hence

$$\frac{\log M_N}{\log N} \longrightarrow 1. \quad (69)$$

Thus no separate growth assumption on M_N is needed.

Set

$$v_N(\sigma) := \frac{J_{P_N}(\sigma)}{\log N}.$$

By Proposition 3.6 and hypothesis (H1),

$$0 \leq v_N(\sigma) \leq \frac{1}{2 \log N} \log E_N(\sigma), \quad (70)$$

so

$$\limsup_{N \rightarrow \infty} v_N(\sigma) \leq (\alpha - \sigma)_+. \quad (71)$$

Fix $\sigma < \alpha$ and choose a compact interval $I \subset (-\infty, \alpha)$ containing σ . Recall from (10) that

$$B_N(\sigma) = \sum_{p \in \mathcal{Q}_N} |a_p|^2 p^{-2\sigma}.$$

For all sufficiently large N , hypothesis (H2) and Proposition 5.1 give

$$v_N(\sigma) \geq \frac{1}{2 \log N} \log B_N(\sigma) - \frac{C_{K_I}}{\log N}. \quad (72)$$

Since the first term tends to $\alpha - \sigma$ and the second tends to zero,

$$\liminf_{N \rightarrow \infty} v_N(\sigma) \geq \alpha - \sigma.$$

Together with (71), this proves

$$v_N(\sigma) \longrightarrow \alpha - \sigma \quad (\sigma < \alpha). \quad (73)$$

For $\sigma \geq \alpha$, (70) and hypothesis (H1) give

$$v_N(\sigma) \longrightarrow 0. \quad (74)$$

Thus v_N converges pointwise on $\mathbb{R}$ to

$$V(\sigma) := (\alpha - \sigma)_+.$$

Each v_N is finite and convex. By Rockafellar [20, Theorem 10.8], pointwise convergence of finite convex functions to the finite convex function V implies uniform convergence on compact subsets. This proves (12).

It remains to pass to the zero-density measures. By Proposition 3.7,

$$\mu_N := \frac{1}{\log M_N} J_{P_N}''$$

is a probability measure. Let $\varphi \in C_c^\infty(\mathbb{R})$. Using the definition of the distributional derivative twice,

$$\begin{aligned} \int_{\mathbb{R}} \varphi(\sigma) d\mu_N(\sigma) &= \frac{1}{\log M_N} \langle J_{P_N}'', \varphi \rangle \\ &= \frac{1}{\log M_N} \langle J_{P_N}, \varphi'' \rangle \\ &= \frac{1}{\log M_N} \int_{\mathbb{R}} J_{P_N}(\sigma) \varphi''(\sigma) d\sigma \\ &= \frac{\log N}{\log M_N} \int_{\mathbb{R}} v_N(\sigma) \varphi''(\sigma) d\sigma. \end{aligned} \quad (75)$$

The support of φ'' is compact, so local uniform convergence of v_N and (69) imply

$$\lim_{N \rightarrow \infty} \int_{\mathbb{R}} \varphi d\mu_N = \int_{\mathbb{R}} V(\sigma) \varphi''(\sigma) d\sigma. \quad (76)$$

Now $V(\sigma) = \alpha - \sigma$ for $\sigma < \alpha$ and $V(\sigma) = 0$ for $\sigma > \alpha$. Let

$$V(\sigma) = (\alpha - \sigma)_+.$$

For every $\varphi \in C_c^\infty(\mathbb{R})$,

$$\langle V'', \varphi \rangle = \langle V, \varphi'' \rangle = \int_{-\infty}^{\alpha} (\alpha - \sigma) \varphi''(\sigma) d\sigma.$$

Integrating by parts gives

$$\int_{-\infty}^{\alpha} (\alpha - \sigma) \varphi''(\sigma) d\sigma = [(\alpha - \sigma) \varphi'(\sigma)]_{-\infty}^{\alpha} + \int_{-\infty}^{\alpha} \varphi'(\sigma) d\sigma.$$

The boundary term vanishes: at $\sigma = \alpha$ the factor $\alpha - \sigma$ is zero, while at $-\infty$ one has $\varphi'(\sigma) = 0$ because φ has compact support. Hence

$$\begin{aligned} \int_{-\infty}^{\alpha} (\alpha - \sigma) \varphi''(\sigma) d\sigma &= \int_{-\infty}^{\alpha} \varphi'(\sigma) d\sigma \\ &= [\varphi(\sigma)]_{-\infty}^{\alpha} \\ &= \varphi(\alpha), \end{aligned}$$

again because $\varphi(\sigma) = 0$ for all sufficiently negative σ . Therefore

$$\langle V'', \varphi \rangle = \varphi(\alpha) = \langle \delta_\alpha, \varphi \rangle,$$

and hence

$$V'' = \delta_\alpha$$

in the sense of distributions, and (76) proves convergence against every test function in $C_c^\infty(\mathbb{R})$. To pass to arbitrary $h \in C_c(\mathbb{R})$, choose $\varphi_j \in C_c^\infty(\mathbb{R})$ with $\|h - \varphi_j\|_\infty \rightarrow 0$. Since both μ_N and δ_α are probability measures,

$$\left| \int (h - \varphi_j) d\mu_N \right| \leq \|h - \varphi_j\|_\infty, \quad |h(\alpha) - \varphi_j(\alpha)| \leq \|h - \varphi_j\|_\infty.$$

First choose j large and then let $N \rightarrow \infty$. It follows that

$$\int_{\mathbb{R}} h d\mu_N \rightarrow h(\alpha) \quad (h \in C_c(\mathbb{R})),$$

which is vague convergence $\mu_N \rightarrow \delta_\alpha$.

For completeness, this vague convergence is in fact weak. Choose $\psi \in C_c(\mathbb{R})$ such that $0 \leq \psi \leq 1$ and $\psi \equiv 1$ on a neighborhood of α . Vague convergence gives

$$\int_{\mathbb{R}} \psi d\mu_N \rightarrow \psi(\alpha) = 1.$$

Since each μ_N is a probability measure,

$$0 \leq \int_{\mathbb{R}} (1 - \psi) d\mu_N = 1 - \int_{\mathbb{R}} \psi d\mu_N \rightarrow 0.$$

Now let g be bounded and continuous. The product $g\psi$ belongs to $C_c(\mathbb{R})$, so vague convergence yields

$$\int_{\mathbb{R}} g\psi d\mu_N \rightarrow g(\alpha)\psi(\alpha) = g(\alpha).$$

Moreover,

$$\left| \int_{\mathbb{R}} g(1 - \psi) d\mu_N \right| \leq \|g\|_\infty \int_{\mathbb{R}} (1 - \psi) d\mu_N \rightarrow 0.$$

Therefore

$$\int_{\mathbb{R}} g d\mu_N \rightarrow g(\alpha),$$

which is precisely weak convergence to δ_α . This proves (13). $\square$

6 The Riemann zeta partial sums

We now verify the hypotheses of Theorem 2.2 in the simplest case.

Proof of Theorem 2.4. Here $a_n = 1$ for $n \leq N$, so $M_N = N$ and

$$E_N(\sigma) = \sum_{n \leq N} n^{-2\sigma}. \quad (77)$$

If $\sigma < 1/2$, the integral test (or Euler summation) gives

$$E_N(\sigma) = \frac{N^{1-2\sigma}}{1-2\sigma} + O_\sigma(1 + N^{-2\sigma}), \quad (78)$$

where the notation O_σ means that the implied constant may depend on σ , but not on N . Since $\sigma < 1/2$,

$$\frac{1}{N^{1-2\sigma}} \longrightarrow 0, \quad \frac{N^{-2\sigma}}{N^{1-2\sigma}} = \frac{1}{N} \longrightarrow 0.$$

Hence

$$E_N(\sigma) = \frac{N^{1-2\sigma}}{1-2\sigma} (1 + o(1)),$$

and therefore

$$\frac{1}{2 \log N} \log E_N(\sigma) \longrightarrow \frac{1}{2} - \sigma.$$

At $\sigma = 1/2$,

$$E_N(1/2) = \log N + O(1),$$

and for $\sigma > 1/2$ the sums are bounded above and below by positive constants. Hence hypothesis (H1) holds with

$$\alpha = \frac{1}{2}.$$

Let

$$\mathcal{P}_N := \{p \text{ prime} : N/2 < p \leq N\}. \quad (79)$$

The prime number theorem gives

$$\#\mathcal{P}_N = \pi(N) - \pi(N/2) = \left(\frac{1}{2} + o(1)\right) \frac{N}{\log N}. \quad (80)$$

Let $I \subset \mathbb{R}$ be compact and put $M_I := \max_{\tau \in I} |\tau|$. If $p, q \in (N/2, N]$ and $\sigma \in I$, then

$$2^{-M_I} \leq \frac{p^{-\sigma}}{q^{-\sigma}} \leq 2^{M_I}. \quad (81)$$

Thus the corresponding coefficients are uniformly comparable. Moreover, for every $p \in (N/2, N]$,

$$2^{-2M_I} N^{-2\sigma} \leq p^{-2\sigma} \leq 2^{2M_I} N^{-2\sigma} \quad (\sigma \in I).$$

Consequently

$$2^{-2M_I} N^{-2\sigma} \#\mathcal{P}_N \leq B_N(\sigma) := \sum_{p \in \mathcal{P}_N} p^{-2\sigma} \leq 2^{2M_I} N^{-2\sigma} \#\mathcal{P}_N. \quad (82)$$

If $I \subset (-\infty, 1/2)$, then (80) and (82) give, uniformly for $\sigma \in I$,

$$\begin{aligned} \frac{1}{2 \log N} \log B_N(\sigma) &= -\sigma + \frac{1}{2 \log N} \log \# \mathcal{P}_N + O_I \left(\frac{1}{\log N} \right) \\ &\longrightarrow \frac{1}{2} - \sigma. \end{aligned}$$

In particular,

$$B_N(\sigma) = N^{1-2\sigma+o(1)} \quad (83)$$

at exponent level for every fixed $\sigma < 1/2$. Taking the abstract set $\mathcal{Q}_N$ in Theorem 2.2 to be $\mathcal{P}_N$, hypothesis (H2) follows. Theorem 2.2 now proves both (14) and (15). $\square$

Combining Theorem 2.4 with Proposition 3.4 gives the direct density formulation.

Corollary 6.1 (Critical-line concentration for ζ_N). *For every $\varepsilon > 0$,*

$$\lim_{N \rightarrow \infty} \frac{2\pi}{\log N} \lim_{T \rightarrow \infty} \frac{1}{2T} \# \{ \rho = \beta + i\gamma : \zeta_N(\rho) = 0, |\beta - 1/2| \geq \varepsilon, |\gamma| < T \} = 0, \quad (84)$$

with the usual convention of avoiding boundary lines carrying zero frequency.

Remark 6.2. Corollary 6.1 does not imply that all zeros approach $\operatorname{Re} s = 1/2$. Mora's theorem [18] shows that, for sufficiently large fixed N , real parts of zeros are dense throughout the admissible critical strip. The two statements are compatible: Mora describes support, whereas (84) describes normalized vertical mass.

7 Dirichlet L -functions

Let χ be a fixed Dirichlet character modulo q . Recall that

$$|\chi(n)|^2 = \mathbf{1}_{(n,q)=1}.$$

We write $\varphi(q) = \#(\mathbb{Z}/q\mathbb{Z})^\times$ for Euler's totient function and μ for the Möbius function, defined by $\mu(1) = 1$, $\mu(n) = (-1)^r$ if n is a product of r distinct primes, and $\mu(n) = 0$ if n is divisible by the square of a prime.

Lemma 7.1. *For fixed q and $\sigma \in \mathbb{R}$,*

$$\sum_{n \leq N} |\chi(n)|^2 n^{-2\sigma} = \sum_{\substack{n \leq N \\ (n,q)=1}} n^{-2\sigma}. \quad (85)$$

If $\sigma < 1/2$, then

$$\sum_{\substack{n \leq N \\ (n,q)=1}} n^{-2\sigma} = \frac{\varphi(q)}{q} \frac{N^{1-2\sigma}}{1-2\sigma} + o_{q,\sigma}(N^{1-2\sigma}), \quad (86)$$

where the notation $o_{q,\sigma}(N^{1-2\sigma})$ means that, for fixed q and σ , the ratio of the error term to $N^{1-2\sigma}$ tends to 0 as $N \rightarrow \infty$.

At $\sigma = 1/2$,

$$\sum_{\substack{n \leq N \\ (n,q)=1}} \frac{1}{n} = \frac{\varphi(q)}{q} \log N + O_q(1), \quad (87)$$

and for $\sigma > 1/2$ the sums remain bounded as $N \rightarrow \infty$.

Proof. Inclusion–exclusion gives

$$\begin{aligned} A_q(x) &:= \sum_{n \leq x} \mathbf{1}_{(n,q)=1} = \sum_{d|q} \mu(d) \left\lfloor \frac{x}{d} \right\rfloor \\ &= x \sum_{d|q} \frac{\mu(d)}{d} + O\left(\sum_{d|q} 1\right) = \frac{\varphi(q)}{q}x + O_q(1). \end{aligned} \quad (88)$$

Set $c_q := \varphi(q)/q$. For $\sigma \neq 0$, Abel summation yields

$$\sum_{\substack{n \leq N \\ (n,q)=1}} n^{-2\sigma} = A_q(N)N^{-2\sigma} + 2\sigma \int_1^N A_q(x)x^{-2\sigma-1} dx. \quad (89)$$

Substituting $A_q(x) = c_q x + O_q(1)$ gives

$$\begin{aligned} \sum_{\substack{n \leq N \\ (n,q)=1}} n^{-2\sigma} &= c_q N^{1-2\sigma} + 2\sigma c_q \int_1^N x^{-2\sigma} dx \\ &\quad + O_q(N^{-2\sigma}) + O_{q,\sigma}\left(\int_1^N x^{-2\sigma-1} dx\right). \end{aligned} \quad (90)$$

Suppose first that $\sigma < 1/2$ and $\sigma \neq 0$. Since

$$\int_1^N x^{-2\sigma} dx = \frac{N^{1-2\sigma} - 1}{1 - 2\sigma},$$

the two main terms in (90) combine to

$$c_q N^{1-2\sigma} + \frac{2\sigma c_q}{1 - 2\sigma} (N^{1-2\sigma} - 1) = \frac{c_q}{1 - 2\sigma} N^{1-2\sigma} + O_{q,\sigma}(1).$$

The remaining terms are $o_{q,\sigma}(N^{1-2\sigma})$: indeed, $N^{-2\sigma}/N^{1-2\sigma} = N^{-1}$, while

$$\int_1^N x^{-2\sigma-1} dx = \begin{cases} O_\sigma(1), & \sigma > 0, \\ O(\log N), & \sigma = 0, \\ O_\sigma(N^{-2\sigma}), & \sigma < 0, \end{cases}$$

and each of these is $o(N^{1-2\sigma})$. The case $\sigma = 0$ follows directly from (88). This proves (86).

At $\sigma = 1/2$, partial summation gives

$$\begin{aligned} \sum_{\substack{n \leq N \\ (n,q)=1}} \frac{1}{n} &= \frac{A_q(N)}{N} + \int_1^N \frac{A_q(x)}{x^2} dx \\ &= c_q + O_q(N^{-1}) + c_q \int_1^N \frac{dx}{x} + O_q\left(\int_1^N \frac{dx}{x^2}\right) \\ &= c_q \log N + O_q(1), \end{aligned}$$

which is (87). Finally, if $\sigma > 1/2$,

$$0 \leq \sum_{\substack{n \leq N \\ (n,q)=1}} n^{-2\sigma} \leq \sum_{n=1}^{\infty} n^{-2\sigma} < \infty,$$

so the partial sums are bounded. $\square$

Proof of Theorem 2.5. Lemma 7.1 verifies hypothesis (H1) with $\alpha = 1/2$.

Let $\mathcal{P}_N$ be the dyadic prime set defined in (79). For $N > 2q$ and $p \in \mathcal{P}_N$, one has $p > q$ and hence $p \nmid q$, so $|\chi(p)| = 1$. Write $\chi(p) = e^{i\vartheta_p}$. Since the rotation $z_p \mapsto e^{i\vartheta_p} z_p$ preserves normalized Haar measure on $\mathbb{T}$, the block

$$\sum_{p \in \mathcal{P}_N} \chi(p) p^{-\sigma} z_p$$

has the same distribution under Haar measure as

$$\sum_{p \in \mathcal{P}_N} p^{-\sigma} z_p.$$

Consequently, for $p, r \in \mathcal{P}_N$ and σ in a compact interval I ,

$$\frac{|\chi(p)| p^{-\sigma}}{|\chi(r)| r^{-\sigma}} = \frac{p^{-\sigma}}{r^{-\sigma}} \leq 2^{\max_{\tau \in I} |\tau|},$$

and

$$\sum_{p \in \mathcal{P}_N} |\chi(p)|^2 p^{-2\sigma} = \sum_{p \in \mathcal{P}_N} p^{-2\sigma} = N^{1-2\sigma+o(1)}$$

locally uniformly for $\sigma < 1/2$, by the zeta calculation. Taking the abstract set $\mathcal{Q}_N$ to be $\mathcal{P}_N$ verifies hypothesis (H2).

Among any q consecutive integers there is an integer congruent to 1 modulo q , hence coprime to q . Therefore

$$N - q + 1 \leq M_{N,\chi} \leq N, \tag{91}$$

and

$$\frac{\log M_{N,\chi}}{\log N} \longrightarrow 1.$$

Theorem 2.2 now gives (16) and (17). $\square$

Remark 7.2. The argument makes no distinction between principal and nonprincipal characters. In particular, the pole of the limiting principal L -function at $s = 1$ has no effect on the vertical mean-motion exponent of its finite truncations.

8 Holomorphic modular forms

Let f be the fixed primitive holomorphic Hecke eigenform introduced in Section 2: it has even weight $k \geq 2$, fixed level Q , trivial Dirichlet character, and no complex multiplication. Its normalized coefficients are $\lambda_f(n) = a_f(n) n^{-(k-1)/2}$.

8.1 Rankin–Selberg quadratic energy

The classical Rankin–Selberg theorem of Rankin [19] and Selberg [24] gives

$$A_f(x) := \sum_{n \leq x} |\lambda_f(n)|^2 = c_f x + O_f(x^{3/5}), \quad c_f > 0. \quad (92)$$

Lemma 8.1 (Weighted Rankin–Selberg energy). *For fixed $\sigma < 1/2$,*

$$\sum_{n \leq N} |\lambda_f(n)|^2 n^{-2\sigma} = \frac{c_f}{1-2\sigma} N^{1-2\sigma} + o_{f,\sigma}(N^{1-2\sigma}). \quad (93)$$

At $\sigma = 1/2$,

$$\sum_{n \leq N} \frac{|\lambda_f(n)|^2}{n} = c_f \log N + O_f(1), \quad (94)$$

and for $\sigma > 1/2$ the weighted partial sums remain bounded.

Proof. For $\sigma \neq 0$, Abel summation gives

$$\sum_{n \leq N} |\lambda_f(n)|^2 n^{-2\sigma} = A_f(N) N^{-2\sigma} + 2\sigma \int_1^N A_f(x) x^{-2\sigma-1} dx. \quad (95)$$

Insert (92). The contribution of the main term $c_f x$ is

$$\begin{aligned} c_f N^{1-2\sigma} + 2\sigma c_f \int_1^N x^{-2\sigma} dx &= c_f N^{1-2\sigma} + \frac{2\sigma c_f}{1-2\sigma} (N^{1-2\sigma} - 1) \\ &= \frac{c_f}{1-2\sigma} N^{1-2\sigma} + O_{f,\sigma}(1). \end{aligned}$$

The contribution of the error $O_f(x^{3/5})$ is

$$O_f(N^{3/5-2\sigma}) + O_{f,\sigma} \left(\int_1^N x^{-2/5-2\sigma} dx \right). \quad (96)$$

The first term divided by $N^{1-2\sigma}$ is $O_f(N^{-2/5})$. For the integral term there are three cases:

$$\int_1^N x^{-2/5-2\sigma} dx = \begin{cases} O_\sigma(N^{3/5-2\sigma}), & \sigma < 3/10, \\ O(\log N), & \sigma = 3/10, \\ O_\sigma(1), & 3/10 < \sigma < 1/2. \end{cases}$$

After division by $N^{1-2\sigma}$ these are respectively $O(N^{-2/5})$, $O(N^{-2/5} \log N)$, and $O(N^{-(1-2\sigma)})$, all tending to zero. This proves (93); the case $\sigma = 0$ follows directly from (92).

At $\sigma = 1/2$, partial summation yields

$$\begin{aligned} \sum_{n \leq N} \frac{|\lambda_f(n)|^2}{n} &= \frac{A_f(N)}{N} + \int_1^N \frac{A_f(x)}{x^2} dx \\ &= c_f + O_f(N^{-2/5}) + c_f \int_1^N \frac{dx}{x} + O_f \left(\int_1^N x^{-7/5} dx \right) \\ &= c_f \log N + O_f(1), \end{aligned}$$

proving (94). Finally, if $\sigma > 1/2$, then $A_f(x) \ll_f x$, and the same partial-summation formula shows that the infinite series $\sum_{n \geq 1} |\lambda_f(n)|^2 n^{-2\sigma}$ converges. $\square$

It follows from Lemma 8.1 that

$$\frac{1}{2 \log N} \log \sum_{n \leq N} |\lambda_f(n)|^2 n^{-2\sigma} \longrightarrow \left(\frac{1}{2} - \sigma \right)_+. \quad (97)$$

8.2 Sato–Tate and isolated prime coordinates

Let $p \nmid Q$ be prime. The Ramanujan–Petersson theorem for holomorphic forms, proved by Deligne [6], gives $|a_f(p)| \leq 2p^{(k-1)/2}$. Because the Dirichlet character is trivial, the normalized Hecke eigenvalue $\lambda_f(p)$ is real. Hence there is an angle $\theta_p \in [0, \pi]$ such that

$$\lambda_f(p) = 2 \cos \theta_p. \quad (98)$$

Since f has no complex multiplication, the Sato–Tate theorem [2, Corollary 7.1.7] states that the angles θ_p , over unramified primes, are equidistributed on $[0, \pi]$ with respect to

$$d\mu_{\text{ST}}(\theta) = \frac{2}{\pi} \sin^2 \theta d\theta. \quad (99)$$

Equivalently, the values $\lambda_f(p) = 2 \cos \theta_p$ are distributed on $[-2, 2]$ according to the semicircle measure

$$d\nu_{\text{ST}}(x) = \frac{1}{2\pi} \sqrt{4 - x^2} dx. \quad (100)$$

Let

$$\mathcal{A} := \{x \in [-2, 2] : 1 \leq |x| \leq 2\}, \quad \eta := \nu_{\text{ST}}(\mathcal{A}) > 0.$$

The boundary of $\mathcal{A}$ has ν_{ST} -measure zero. Therefore [2, Corollary 7.1.7] gives

$$\#\{p \leq x : p \nmid Q, 1 \leq |\lambda_f(p)| \leq 2\} = \eta \pi(x) + o(\pi(x)). \quad (101)$$

Applying this at $x = N$ and $x = N/2$, and using the prime number theorem, yields

$$\#\{N/2 < p \leq N : p \nmid Q, 1 \leq |\lambda_f(p)| \leq 2\} = \left(\frac{\eta}{2} + o(1) \right) \frac{N}{\log N}. \quad (102)$$

Define

$$\mathcal{P}_{N,f} := \{p : N/2 < p \leq N, p \nmid Q, 1 \leq |\lambda_f(p)| \leq 2\}. \quad (103)$$

Then $\#\mathcal{P}_{N,f} \rightarrow \infty$.

Let $I \subset \mathbb{R}$ be compact and put $M_I := \max_{\tau \in I} |\tau|$. If $p, r \in \mathcal{P}_{N,f}$ and $\sigma \in I$, then $1 \leq |\lambda_f(p)|, |\lambda_f(r)| \leq 2$ and $1/2 < p/r < 2$. Hence

$$\frac{1}{2^{M_I+1}} \leq \frac{|\lambda_f(p)| p^{-\sigma}}{|\lambda_f(r)| r^{-\sigma}} \leq 2^{M_I+1}. \quad (104)$$

Thus the isolated weights are uniformly comparable. Moreover, for $\sigma < 1/2$,

$$B_{N,f}(\sigma) := \sum_{p \in \mathcal{P}_{N,f}} |\lambda_f(p)|^2 p^{-2\sigma}. \quad (105)$$

Since $1 \leq |\lambda_f(p)|^2 \leq 4$ and $N/2 < p \leq N$,

$$2^{-2|\sigma|} N^{-2\sigma} \#\mathcal{P}_{N,f} \ll_{\sigma} B_{N,f}(\sigma) \ll_{\sigma} N^{-2\sigma} \#\mathcal{P}_{N,f}.$$

Together with (102), this gives

$$B_{N,f}(\sigma) = N^{1-2\sigma+o(1)}, \quad (106)$$

with the $o(1)$ uniform for σ in compact subsets of $(-\infty, 1/2)$. Hence

$$\frac{1}{2 \log N} \log B_{N,f}(\sigma) \longrightarrow \frac{1}{2} - \sigma \quad (107)$$

locally uniformly on $(-\infty, 1/2)$.

8.3 Proof of the modular theorem

Proof of Theorem 2.6. Equation (97) verifies hypothesis (H1) with $\alpha = 1/2$. Taking the abstract set $\mathcal{Q}_N$ to be $\mathcal{P}_{N,f}$, equations (104) and (107) verify hypothesis (H2).

By (102), for all sufficiently large N there is a prime $p \in (N/2, N]$ with $\lambda_f(p) \neq 0$. Consequently

$$N/2 < M_{N,f} \leq N, \quad \frac{\log M_{N,f}}{\log N} \longrightarrow 1.$$

Theorem 2.2 now gives (21) and (22).

Finally, from $a_f(n) = \lambda_f(n)n^{(k-1)/2}$,

$$\begin{aligned} \mathcal{L}_N(s, f) &= \sum_{n \leq N} a_f(n) n^{-s} \\ &= \sum_{n \leq N} \lambda_f(n) n^{-(s-(k-1)/2)} \\ &= L_N \left(s - \frac{k-1}{2}, f \right). \end{aligned} \quad (108)$$

Thus the real parts of all zeros are translated by $(k-1)/2$, and the concentration line $\operatorname{Re} w = 1/2$ in the normalized variable $w = s - (k-1)/2$ becomes

$$\operatorname{Re} s = \frac{1}{2} + \frac{k-1}{2} = \frac{k}{2}.$$

□

Corollary 8.2. *In particular, Theorem 2.6 applies to every normalized holomorphic Hecke cusp eigenform for $\mathrm{SL}_2(\mathbb{Z})$. Zero-density concentration occurs on $\operatorname{Re} s = 1/2$ for the normalized Hecke-eigenvalue truncations and on $\operatorname{Re} s = k/2$ for the classical Fourier-coefficient truncations.*

9 Conclusion

We have proved a general criterion that converts quadratic coefficient growth, together with a sufficiently large family of isolated prime coordinates, into concentration of normalized vertical zero density on a single line. The Bohr–Jessen identity turns the vertical logarithmic mean into a torus Mahler measure, while the translation-uniform Steinhaus estimate prevents the isolated linear block from spending excessive Haar measure near zero. The resulting lower bound matches the elementary L^2 upper bound at logarithmic scale.

The limiting potential is

$$V_\alpha(\sigma) = (\alpha - \sigma)_+,$$

whose unique corner gives $V_\alpha'' = \delta_\alpha$. For the partial sums of the Riemann zeta function and of fixed Dirichlet L -functions, $\alpha = 1/2$. Rankin–Selberg and Sato–Tate give the same value for normalized holomorphic Hecke eigenforms of fixed level without complex multiplication; in the classical Fourier-coefficient normalization the corresponding line is $\operatorname{Re} s = k/2$.

The present argument is intrinsically formulated in the Jessen mean-motion regime, with $T \rightarrow \infty$ for each fixed N before $N \rightarrow \infty$. A coupled (N, T) limit would require quantitative control of the passage from vertical averages to Haar averages that is uniform in the growing frequency set $\{\log p : p \leq N\}$, and no such uniform estimate is used here.

The conclusion is compatible with the much wider support of individual zeros for a fixed truncation: concentration of normalized vertical density is a measure-theoretic statement and does not assert geometric concentration of the entire zero set.

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